CityHelp

A platform for receiving, analyzing, moderating, and processing citizen appeals

Languages: Русский | English

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What This Project Is

CityHelp is a city-focused digital system for handling citizen appeals.

The platform combines:

- appeal creation with a map, photos, and description
- automatic AI-based analysis
- manual moderation by an administrator
- employee-side execution workflow
- final quality review
- user rating of the final result
- dashboard analytics for appeal monitoring

CityHelp is designed as a single point of interaction between residents, employees, and administration.

Core Product Goals

- make the appeal journey transparent
- reduce manual routine through AI-assisted analysis
- improve classification and routing speed
- keep the full history of each appeal in one place
- control quality through moderation and rating
- provide operational analytics for administration

System Roles

Role	Responsibility
`User`	Creates an appeal, uploads photos, marks a point on the map, tracks statuses, rates the result

Role	Responsibility
`Employee`	Receives assigned appeals, accepts them into work, uploads result photos and leaves comments
`Admin`	Reviews AI analysis, approves or rejects an appeal, changes assignee, sends work for revision
`SuperAdmin`	Can do everything an `Admin` can do and has full system-level control

Technology Stack

Layer	Technologies	Purpose
Frontend	`Nuxt 4` , `Vue 3` , `SCSS`	Interface, pages, components, SSR/SPA logic
State management	`Pinia`	Global state for user data, API state, notifications, loaders, and UI logic
Backend	`Nitro` , `Node.js` , `h3`	API routes, server logic, middleware, request handling
Database	`MongoDB` , `Mongoose`	Storage for appeals, users, categories, FAQ, and AI prompts
AI	`Gemini API`	Appeal photo/text analysis, classification, priority, summaries, and evidence
Maps	`Yandex Maps API` , `Heatmap module`	Point selection, reverse geocoding, and dashboard heatmap
Storage	`Vercel Blob`	Storage for appeal photos, result photos, and avatars
i18n	`@nuxtjs/i18n`	Localization for `kz` , `ru` , and `en`
UI helpers	`maska` , `nuxt-swiper` , `vue3-tel-input`	Input masks, UI helpers, and phone inputs

Layer	Technologies	Purpose
Hosting	`Vercel`	Production deployment, serverless API hosting, preview workflow
Team tools	`GitHub` , `Trello` , `Figma`	Version control, task tracking, and design/product collaboration

Interface Languages

- `kz` - default locale
- `ru`
- `en`

Project Architecture

CityHelp is built as a unified Nuxt fullstack project where the client and server live in the same repository.

Top-Level Structure

Folder	Purpose
`app/`	Client side: pages, components, composables, stores, layouts
`server/`	Backend on Nitro: API, controllers, services, models, utilities
`i18n/`	Localization config and dictionaries
`public/`	Static assets, images, icons, and fonts

Backend Structure

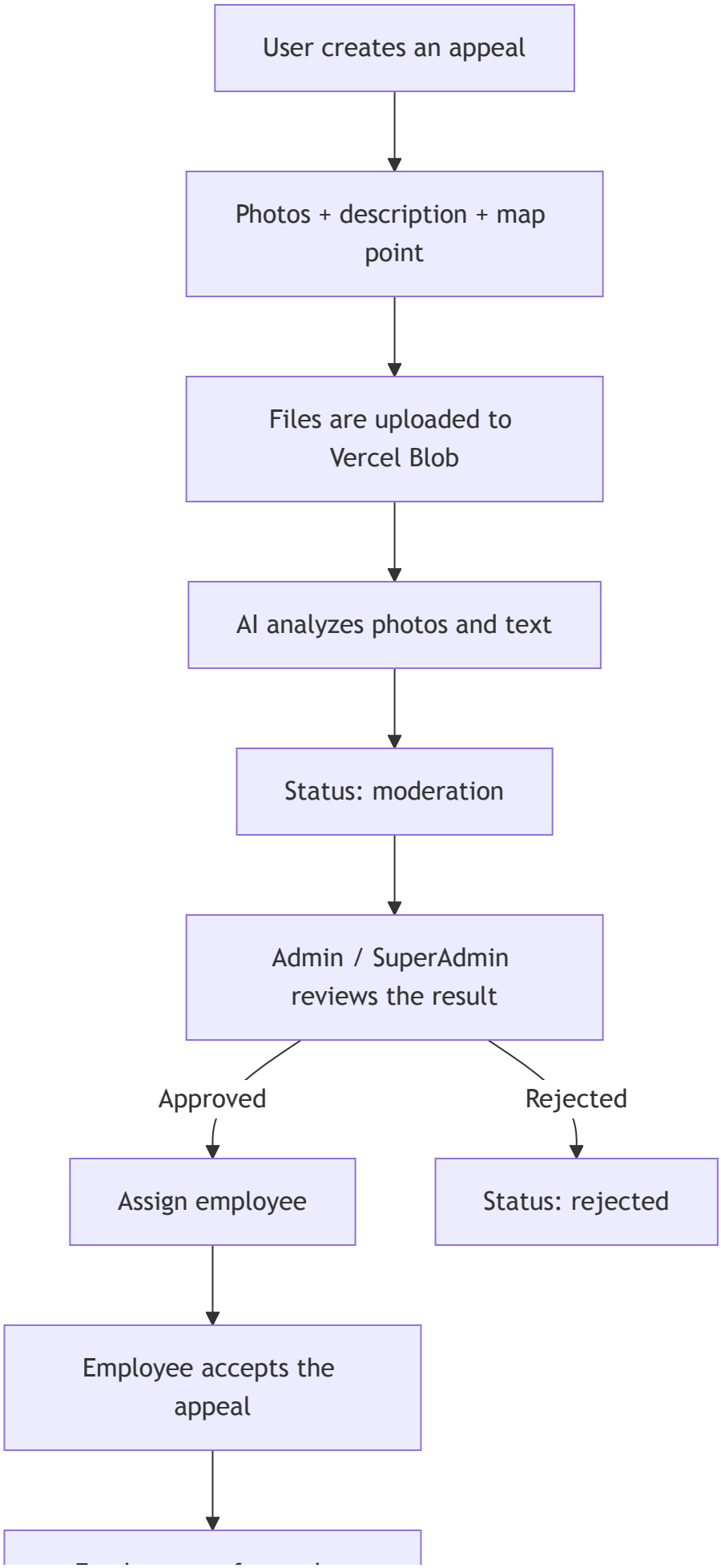
Folder	Purpose
`server/api/`	API entry points
`server/controllers/`	Request/response orchestration
`server/services/`	Business logic

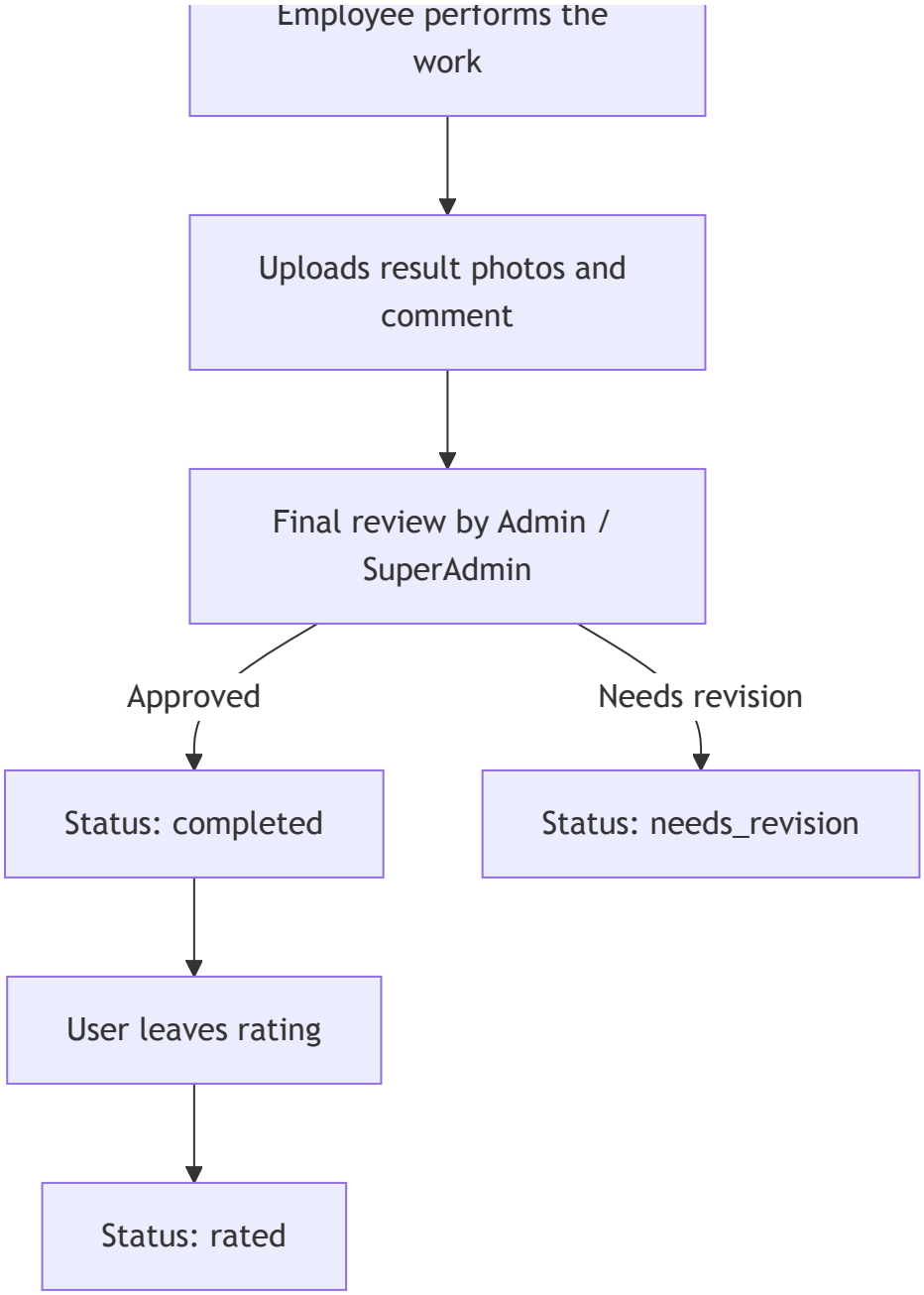
Folder	Purpose
`server/models/`	Mongoose schemas and models
`server/config/`	Infrastructure configuration such as MongoDB
`server/utils/`	Shared utilities

Why This Approach Works Well

- frontend and backend stay synchronized in one codebase
- role logic, API logic, and UI logic are easier to manage together
- deployment to Vercel is simpler as a single system
- shared entities and product logic are easier to maintain

Appeal Flow


User creates an appeal
Photos + description + map point
Files are uploaded to Vercel Blob
AI analyzes photos and text
Status: moderation
Admin / SuperAdmin reviews the result
Approved
Rejected
Assign employee
Status: rejected
Employee accepts the appeal



Status Lifecycle

Status	Meaning	Who changes it
`new`	New confirmed appeal	Admin / SuperAdmin
`moderation`	AI already processed the appeal, waiting for human review	System + Admin review
`processing`	Appeal is accepted and currently in work	Employee
`needs_revision`	The submitted result requires improvement	Admin / SuperAdmin
`completed`	Work is accepted as completed	Admin / SuperAdmin

Status	Meaning	Who changes it
`rated`	User has rated the result	User
`rejected`	Appeal has been rejected	Admin / SuperAdmin

Priorities And Deadlines

Priority	Meaning	Base deadline
`low`	Low	7 days
`medium`	Medium	4 days
`high`	High	3 days
`urgent`	Urgent	1 day

In the current pipeline, the deadline is calculated on the server based on priority rather than directly by Gemini.

AI Pipeline

In CityHelp, AI does more than simple classification. It prepares a structured analysis for the moderator.

What AI Receives

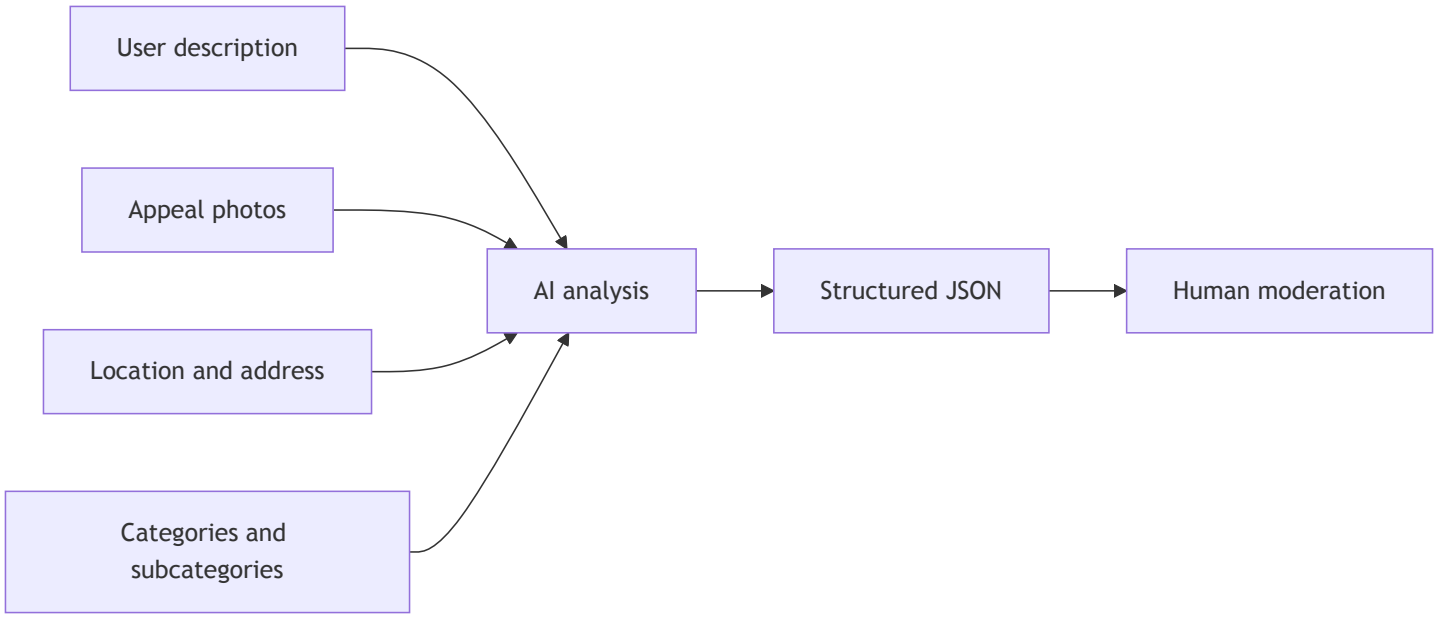
- appeal text
- appeal photos
- coordinates and address
- available categories and subcategories
- category reference context

What AI Produces

Field	Meaning
`photoObservation`	Short description of what is visible in the image

Field	Meaning
`photoDetails`	Detailed visual findings
`shortSummary`	Short summary of the issue
`analysisSummary`	Explanation of the analysis
`category`	Main category
`subCategory`	Subcategory
`priority`	Appeal priority
`evidence`	Facts supporting the result
`uncertainties`	What remains unclear
`needsClarification`	Whether clarification is needed
`categoryReason`	Why this category was chosen
`subCategoryReason`	Why this subcategory was chosen
`priorityReason`	Why this priority was chosen

Pipeline Structure



Important Notes

- `Gemini` is used with model fallback logic
- AI responds in JSON format
- prompts can be stored and edited in the database
- there is a dedicated admin UI for `AI prompts`
- the final moderation decision always remains with a human

Data And Models

Main entities in MongoDB:

Entity	Purpose
`Appeal`	Appeal record, statuses, location, photos, AI result, timeline, rating
`User`	User, employee, admin, or superadmin
`Category`	Appeal categories and subcategories
`Faq`	FAQ knowledge base
`Prompt`	AI prompts and versions
`AiTrainingCase`	Cases used to review and improve AI quality

What `Appeal` Stores

- appeal author
- issue description
- up to 5 user photos
- coordinates and address
- category and subcategory
- priority and deadline
- assigned employee
- moderation and execution notes
- result photos
- `aiResult`
- `timeline` of the full appeal journey

- user `rating`

Why MongoDB Fits This Product

- nested structures like `location`, `timeline`, `rating`, and `aiResult` are easy to store
- schemas can evolve without heavy relational migrations
- it works well for the document-oriented shape of an appeal
- Mongoose provides validation, indexes, and model-level structure

Maps, Geodata, And Analytics

Why Yandex Maps Is Used

- the user selects the appeal location on the map
- the system saves `x/y` coordinates
- reverse geocoding is used to resolve the address
- the appeal is stored together with location data

Heatmap

There is a heatmap on the dashboard.

It is used to:

- identify hot spots across the city
- quickly understand issue concentration
- support dashboard analytics for administration

Authorization And Access

The project includes registration, login, logout, and role-based access control.

What Is Used

- the token is generated on the server
- the token is stored in a cookie

- the user role is included in the payload
- middleware and API routes verify permissions for routes and actions

Why It Matters

- separates user flows from admin flows
- provides role-based access to actions
- protects moderation, staff, and admin tools

File Storage

`Vercel Blob` is used for images and file storage.

Base Storage Structure

File type	Path
User photos	`cityhelp/appeals/<appealId>/photos/*`
Result photos	`cityhelp/appeals/<appealId>/fixed-images/*`
Avatars	`cityhelp/avatars/<userId>/*`

Why It Is Convenient

- each file set is tied to a specific entity
- it is easier to delete all appeal-related files when needed
- the structure is predictable and scalable
- storage stays cleaner in production

Deployment And Operations

Production

- production deployment is hosted on `Vercel`
- backend functionality runs through Nuxt/Nitro server API routes

- images and attachments are stored in `Vercel Blob`

Operational Notes

- production build and API behavior can be verified in one environment
- Vercel logs are useful for diagnosing backend behavior
- API errors return `statusCode` and `statusMessage`, which helps debugging

Runtime Configuration

The project uses runtime configuration for:

- MongoDB
- auth secret
- Gemini API
- base API URL

Secrets should be stored through environment variables and Vercel project settings.

Team Workflow And Project Management

Tooling

Tool	How it is used
`Trello`	Task tracking, priorities, weekly planning, backlog, bugs, and test workflow
`GitHub`	Repository, change history, branching, and collaboration
`Figma`	UI structure, materials, and design/product discussions
`Vercel`	Production deployment and deployment verification

Typical Workflow

1. a task is created in `Trello`
2. structure and priority are discussed
3. implementation is done in code and tracked in `GitHub`
4. the feature is deployed through `Vercel`

5. production behavior and logs are checked if needed

Why This Works Well

- priorities stay visible
- task status is easy to understand
- work is distributed clearly between contributors
- the path from idea to deployment stays fast

Current Project Status

The status below is based on the project board and is current as of **April 14, 2026**.

Completed

Done: 06.04.26 - 12.04.26

- create appeal flow and main API flow
- connect `Gemini AI` to the server
- implement `Login / Register API`
- connect AI token flow

Done: 13.04.26 - 19.04.26

- user profile
- AI training for more accurate analysis
- discussion and clarification of all status types
- priority-based deadlines
- database structure

Currently In Progress

- materials for the landing page

Planned For The Upcoming Week

- `Employee` department / division
- user -> employee system

- docs API

In Testing And Backlog

- testing
- full translation
- multi-city split

Postponed Or Rejected

- corporate email `info@cityhelp.com / .kz`
- moving the domain from `PS.kz` to a separate server
- `2GIS API` integration

Main Pages

Page	Purpose
`/`	Landing page
`/panel`	Main analytics dashboard
`/panel/create-appeal`	Appeal creation
`/panel/appeal/:id`	Appeal detail page
`/panel/edit-appeal/:id`	Appeal editing during moderation
`/panel/user/my-appeals`	User appeal history
`/panel/employee/appeals`	Employee queue
`/panel/admin/appeals`	Main admin appeal list
`/panel/admin/users`	User management
`/panel/admin/staff`	Staff list and statistics
`/panel/admin/categories`	Category management
`/panel/admin/faq`	FAQ management

Page	Purpose
`/panel/admin/prompts`	AI prompt management
`/panel/admin/translations`	Translation management

Local Setup

Requirements

- `Node.js`
- `npm`
- MongoDB access
- configured runtime secrets for AI and storage

Commands

```
npm install
npm run dev
```

Scripts

```
npm run dev
npm run build
npm run generate
npm run preview
```

Resources

Resource	Link
Production	https://cityhelp-diploma-yij7.vercel.app
GitHub	https://github.com/yenlikksarybay/cityhelp-diploma

Resource	Link
Trello board	https://trello.com/b/aotyaf6W/cityhelp-diploma

Summary

CityHelp is not just a complaint form. It is a full appeal-processing lifecycle:

1. the user creates an appeal
2. the system stores location and files
3. AI performs the initial structured analysis
4. administration reviews and routes the appeal
5. the employee performs the work
6. administration verifies the result
7. the user rates the outcome

This approach makes the process:

- transparent
- manageable
- scalable
- understandable for the user
- practical for employees
- controllable for administration